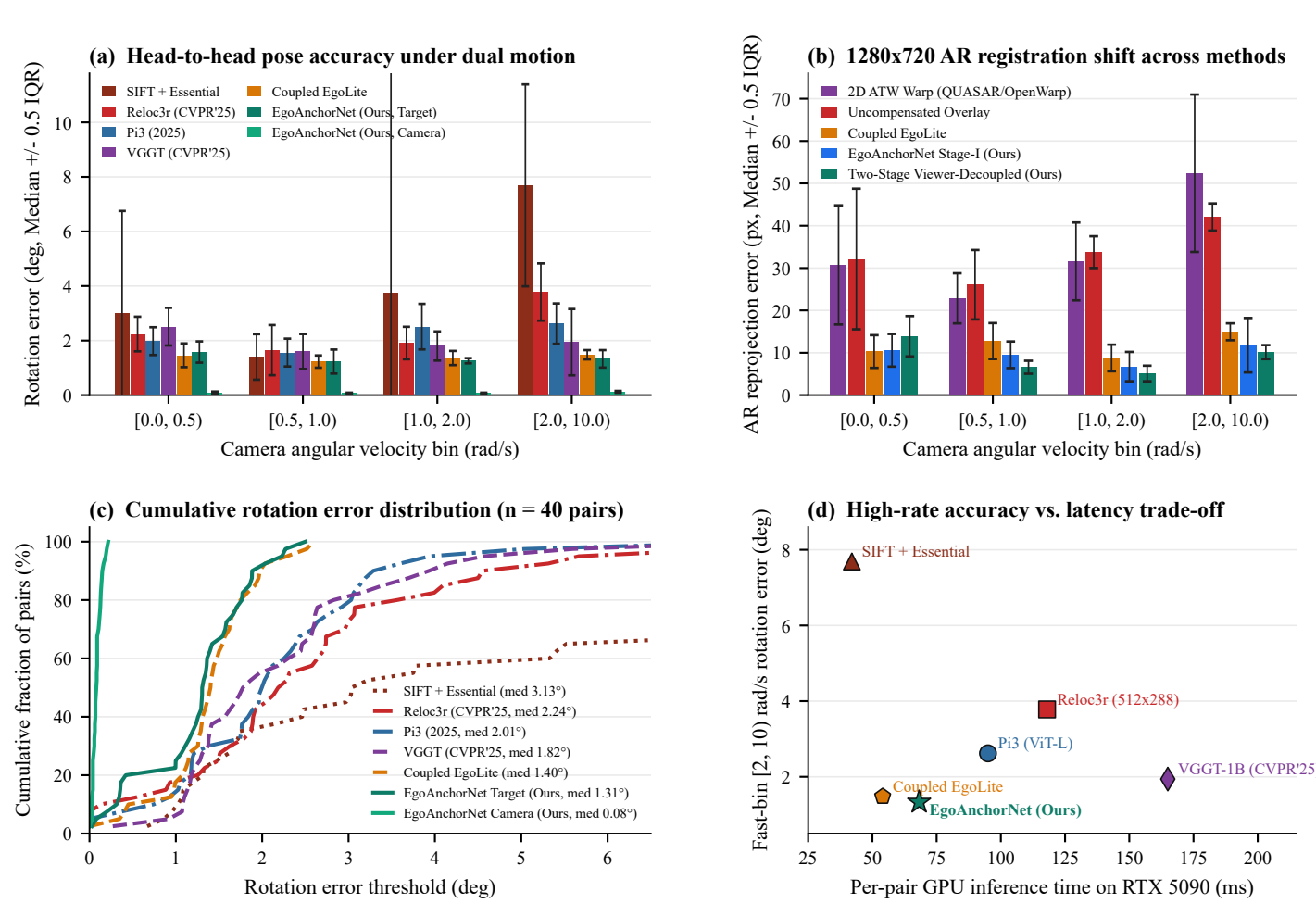

(a) Head-to-head pose accuracy under dual motion
SIFT + Essential
Reloc3r (CVPR'25)
Pi3 (2025)
VGGT (CVPR'25)
Coupled EgoLite
EgoAnchorNet (Ours, Target)
EgoAnchorNet (Ours, Camera)
Rotation error (deg, Median +/- 0.5 IQR)
[0.0, 0.5)
[0.5, 1.0)
[1.0, 2.0)
[2.0, 10.0)
Camera angular velocity bin (rad/s)
(b) 1280x720 AR registration shift across methods
2D ATW Warp (QUASAR/OpenWarp)
Uncompensated Overlay
Coupled EgoLite
EgoAnchorNet Stage-I (Ours)
Two-Stage Viewer-Decoupled (Ours)
AR reprojection error (px, Median +/- 0.5 IQR)
Camera angular velocity bin (rad/s)
(c) Cumulative rotation error distribution (n = 40 pairs)
Cumulative fraction of pairs (%)
SIFT + Essential (med 3.13°)
Reloc3r (CVPR'25, med 2.24°)
Pi3 (2025, med 2.01°)
VGGT (CVPR'25, med 1.82°)
Coupled EgoLite (med 1.40°)
EgoAnchorNet Target (Ours, med 1.31°)
EgoAnchorNet Camera (Ours, med 0.08°)
Rotation error threshold (deg)
(d) High-rate accuracy vs. latency trade-off
Fast-bin [2, 10) rad/s rotation error (deg)
SIFT + Essential
Reloc3r (512x288)
Pi3 (ViT-L)
VGGT-1B (CVPR'25
Coupled EgoLite
EgoAnchorNet (Ours)
Per-pair GPU inference time on RTX 5090 (ms)